

KUO-HAN HUNG

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Education

Stanford University

M.S. in Computer Science

June 2027 (Expected)

Stanford, CA

- GPA: 4.0/4.0; Specialization: Artificial Intelligence and Robotics.

National Taiwan University

B.S. in Computer Science and Information Engineering

Dec. 2024

Taipei, Taiwan

- GPA: 4.24/4.3.

Selected Publications

(* indicates equal contribution)

[1] Reinforcement Learning for Real-Time Vision-Language-Action Policies

Perry Dong*, **Kuo-Han Hung***, Dorsa Sadigh, and Chelsea Finn
Under Review

[2] EXPO-FT: Sample-Efficient Reinforcement Learning Finetuning for Vision-Language-Action Models

Perry Dong*, **Kuo-Han Hung***, Tian Gao, Dorsa Sadigh, and Chelsea Finn
Conference on Robot Learning (CoRL), 2026

[3] TQL: Scaling Q-Functions with Transformers by Preventing Attention Collapse

Perry Dong*, **Kuo-Han Hung***, Alexander Swerdlow, Dorsa Sadigh, and Chelsea Finn
International Conference on Machine Learning (ICML), 2026

[4] VICtoR: Learning Hierarchical Vision-Instruction Rewards for Long-horizon Manipulation

Kuo-Han Hung*, Pang-Chi Lo*, Jia-Fong Yeh*, Han-Yuan Hsu, Yi-Ting Chen, and Winston Hsu
International Conference on Learning Representations (ICLR), 2025

[5] Learning Skill Abstraction from Action-Free Videos via Optical Flow

Hung-Chieh Fang*, **Kuo-Han Hung***, Chu-Rong Chen, . . . , Min-Hung Chen, and Shao-Hua Sun
ICML Workshop on Building Physically Plausible World Models, 2025

[6] Attention Tracker: Detecting Prompt Injection Attacks in LLMs

Kuo-Han Hung, Ching-Yun Ko, Ambrish Rawat, I-Hsin Chung, Winston Hsu, and Pin-Yu Chen
Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL), 2025

[7] AED: Adaptable Error Detection for Few-shot Imitation Policy

Jia-Fong Yeh, **Kuo-Han Hung**, Pang-Chi Lo, Chi-Ming Chung, . . . , Yi-Ting Chen, and Winston Hsu
Conference on Neural Information Processing Systems (NeurIPS), 2024

Research Experience

Stanford Artificial Intelligence Laboratory (ILIAD & IRIS Labs)

Sept. 2025 – Present

Graduate Researcher, Advisors: Prof. Dorsa Sadigh & Prof. Chelsea Finn

Stanford, CA

- **Visual representation pretraining** for reinforcement learning (ongoing).
- **RL finetuning of real-time VLA policies** under inference latency. [1]
 - Decoupled slow VLA action proposals from a fast edit policy that reacts to the latest observation.
 - Raised success rate from 42% to 97% on four dynamic real-world tasks with 10 min of robot data.
- **Sample-efficient RL finetuning of pretrained VLAs** in the real world. [2]
 - Built a stable RL finetuning system with residual action edits and Q-guided sampling.
 - Reached 30/30 success on all tasks with 19 min of online data on average; code open-sourced.

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- **Scaling Q-functions** with Transformers in offline RL. [3]
 - Introduced attention entropy regularization to stabilize and scale Transformer value functions.
 - Showed that prior methods fail to scale, while TQL gains 43% as model size grows.

Embodied AI Lab

June 2025 – Sept. 2025

Undergraduate Researcher, Advisor: Prof. Tsung-Wei Ke

Taipei, Taiwan

- **Generalized dexterous manipulation policies** via digital twins.
 - Built a pipeline to convert videos into digital twins using mesh generation and object/human pose estimation.
 - Trained residual RL policies on digital twins from in-the-wild videos, yielding diverse dexterous behaviors.

Robot Learning Lab

Oct. 2024 – May 2025

Undergraduate Researcher, Advisor: Prof. Shao-Hua Sun

Taipei, Taiwan

- **Manipulation policies learned from action-free videos.** [5]
 - Designed a Transformer to learn skills from action-free videos, using optical flow as an action surrogate.
 - Enabled multi-task and long-horizon planning by translating flow-based skill plans into executable actions.

Communication and Multimedia Lab

May 2022 – May 2024

Undergraduate Researcher, Advisors: Prof. Winston H. Hsu & Prof. Yi-Ting Chen

Taipei, Taiwan

- **Vision-instruction correlation rewards** for long-horizon manipulation. [4]
 - Finetuned LLMs and VLMs into a hierarchical video reward model for generating stage-aware signals.
 - Achieved a 43% improvement in success rate over state-of-the-art baselines on long-horizon tasks.
- **Behavior error detection in few-shot imitation policies** via representation pattern analysis. [7]

Work Experience

IBM Research

June 2024 – Sept. 2024

Research Intern, Host: Dr. Pin-Yu Chen

Yorktown Heights, NY

- Found that attention drifts from the original instruction during **prompt injection attacks on LLM agents.** [6]
- Traced the effect to specific attention heads using mechanistic interpretability.
- Designed an attention-based detector, a 31% AUROC improvement over the prior state of the art.

Microsoft

July 2023 – June 2024

Applied Scientist Intern, Host: Dr. Chiqun Zhang

Mountain View, CA (Remote)

- **Integrated Bing Maps with Microsoft Copilot** for conversational, multi-destination map queries.
- Formulated a pathfinding algorithm that learns from history, **achieving a 10× speedup over Dijkstra.**
- Developed an LLM-powered search system that adapts to user intent and geographic context.

Teaching Experience

CS 5043: Machine Learning (graduate level)

Spring 2023

Teaching Assistant, National Taiwan University

Taipei, Taiwan

- Co-designed homework on generalization theory and algorithm analysis (e.g. SVM, Random Forest, VC dimension) and a final project on ordinal ranking, for a class of 250+ students.

Awards & Honors

- **Bachelor Thesis Dean's Award**, National Taiwan University — top 2 theses in the department. May 2024
- **Presidential Award**, National Taiwan University — top 2% of the academic year. Sept. 2022, Sept. 2023
- **Dean's List**, National Taiwan University — top 5% of the academic year. June 2021, 2022, 2023

Technical Skills

- **Languages:** Python, C/C++, Golang, JavaScript, SQL, Bash.
- **Research Areas:** Reinforcement Learning, Vision-Language-Action Models, LLMs & VLMs, Diffusion Models, Mechanistic Interpretability.
- **ML Toolkits:** PyTorch, JAX, NumPy, Scikit-Learn, Pandas, Matplotlib.
- **Infrastructure & Other:** Linux, Docker, Git, ROS, Azure, React, MongoDB.

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